

PRODUCT REQUIREMENTS DOCUMENT (PRD)

AI Multi-Agent Life & Career Copilot

Version 2.0 — Web + Mobile Ready

Backend: Python + Django + Django REST Framework

Web: React.js

Mobile: React Native

Database: PostgreSQL

1. Executive Summary

AI Multi-Agent Life & Career Copilot is an AI-powered career platform for students, freshers and job seekers. A Supervisor Agent coordinates specialized AI agents for career analysis, skill gaps, learning roadmaps, projects, resumes, job matching, interviews, RAG and AI flashcards. The product is API-first so one Django backend serves both the React website and the future React Native Android/iOS application.

2. Vision & Goals

Build a personal AI career team that guides users from career discovery through skill development, portfolio building, job matching, interview preparation and continuous revision.

- Personalized career recommendations
- Prioritized skill-gap analysis
- Adaptive learning roadmaps
- Career-relevant project recommendations
- Resume optimization
- Job matching
- AI interview practice
- Interactive AI flashcards
- RAG-based document assistance
- Progress tracking
- Web and mobile access through one backend

3. Problem Statement

- Career choices are often unclear or generic.
- Students cannot easily identify role-specific skill gaps.
- Learning resources and plans are scattered.
- Project selection is often disconnected from career goals.
- Resumes may not match job descriptions.
- Interview preparation lacks personalized evaluation.
- Revision is inefficient without structured practice.

4. Target Users

- College students
- Final-year students
- Freshers
- Job seekers

- Career switchers

5. User Roles

Student

- Register/login
- Create profile
- Set career goals
- Manage skills
- Upload resumes/documents
- Use AI career features
- Practice interviews
- Generate/review flashcards
- Track progress

Admin

- Manage users
- Monitor AI usage
- Manage reference career/job data
- Review system logs
- Manage platform data through Django Admin

6. Platform Architecture

React.js Web App ■■

■→ Django REST API → AI Agent System → PostgreSQL

React Native App ■■■■ ↓

RAG + Vector DB + LLM

Django is the single source of truth for authentication, business logic, AI orchestration, database access and progress. The clients consume versioned APIs.

7. Technology Stack

Layer	Technology
Web	React.js
Mobile	React Native
Backend	Python + Django + Django REST Framework
Auth	Django Auth + JWT
Database	PostgreSQL
AI orchestration	LangGraph / LangChain
LLM	Configurable LLM API
RAG	Embeddings + ChromaDB initially; Qdrant when needed
Documents	PyPDF + python-docx
API	REST, versioned /api/v1/

8. Multi-Agent System

Supervisor Agent

Understands the request, selects relevant agents, assigns tasks, combines outputs and sends results for verification.

Career Analysis Agent

Analyzes education, interests, skills, projects and goals to recommend career paths and specializations.

Skill Gap Agent

Compares current skills against target-role requirements and prioritizes missing skills.

Learning Roadmap Agent

Creates an adaptive timeline based on skill gaps, current knowledge, target role and available time.

Project Recommendation Agent

Recommends projects aligned with career goals and missing skills, including stack, features, difficulty and milestones.

Resume Agent

Extracts and analyzes resume content and provides target-role-specific improvements.

Job Matching Agent

Compares profile/resume against job descriptions and returns a match estimate, matching skills and missing skills.

Interview Agent

Generates role-specific questions, evaluates answers and gives feedback and scores.

Flashcard AI

Generates question-answer cards from topics, text, notes or documents and tracks review progress.

Verification Agent

Checks outputs for contradictions, missing context and weak recommendations before final presentation.

9. Career Analysis

- Career goal selection
- Profile-based analysis
- Multiple suitable paths where appropriate
- Explanation for recommendations
- Required skills
- Personalized next steps

10. Skill Gap Analysis

- Current skill inventory
- Target-role requirements
- Skill comparison
- Gap severity/priority
- Recommended learning order
- Completion tracking

11. Learning Roadmap

- Timeline
- Stages and topics
- Priority
- Practice tasks

- Completion tracking
- Roadmap regeneration when goals or skills change

12. Projects

- Project title
- Description
- Difficulty
- Required skills
- Technology stack
- Features
- Milestones
- Portfolio value

13. Resume Analyzer

- PDF/DOCX upload
- Secure validation
- Text extraction
- Skill/project/education extraction
- Target-role comparison
- Keyword suggestions
- Improvement recommendations
- Estimated relevance score

Any ATS-style score must be clearly labeled as an AI-generated estimate, not an official ATS score.

14. Job Matching

- Job description input/upload
- Skill comparison
- Matching skills
- Missing skills
- Match estimate
- Improvement recommendations
- Save job matches

15. AI Interview

- Role selection
- Difficulty selection
- AI questions
- Answer submission
- Technical evaluation
- Clarity/completeness feedback
- Score tracking

- Interview history

Voice interview is a future enhancement, not an MVP dependency.

16. AI Flashcards

Users can generate cards from a topic, text, notes or PDF. The front shows the question and the back shows the answer.

- Tap/click to flip
- Flip animation
- Next/previous
- Shuffle
- Mark difficult
- Review difficult cards
- Save progress
- Future spaced repetition

Progress is stored on the backend so web and mobile can share the same decks and review history.

17. RAG

Document Upload → Text Extraction → Chunking → Embeddings → Vector Database → Retrieval → LLM → Contextual Response

- Resume/document upload
- Semantic search
- Document-aware AI chat
- Context-based answers
- Source/context handling

18. AI Chat

- Career questions
- Profile-aware responses
- Roadmap-aware responses
- Document-aware responses via RAG
- Conversation history
- Uncertainty/error handling

19. Dashboard & Progress

- Career goal
- Career readiness estimate
- Skill progress
- Roadmap progress
- Project progress
- Interview performance
- Flashcard activity
- Recent recommendations

- Next recommended action

20. Web Pages

- Home
- About
- Features
- Login
- Register
- Dashboard
- Profile
- Career Analysis
- Skill Gap
- Learning Roadmap
- Projects
- Resume Analyzer
- Job Matching
- Interview Practice
- AI Flashcards
- AI Chat
- Progress
- Settings

21. Mobile Pages

- Onboarding
- Login/Register
- Dashboard
- Profile
- Career Analysis
- Skill Gap
- Learning Roadmap
- Projects
- Resume Analyzer
- Job Matching
- Interview Practice
- AI Flashcards
- AI Chat
- Progress
- Settings
- Notifications

React Native will consume the same Django APIs. Mobile-specific features can include push notifications, camera/document scanning and microphone-based interview practice.

22. API-First Requirements

- REST APIs for all core modules
- API versioning: /api/v1/
- JWT/token authentication
- Server-side permissions
- Pagination
- Consistent JSON responses/errors
- Validated file uploads
- Request IDs/status for long AI tasks
- No duplicated business logic in React clients

23. Suggested API Modules

/api/v1/auth/
/api/v1/profile/
/api/v1/career/
/api/v1/skills/
/api/v1/roadmap/
/api/v1/projects/
/api/v1/resumes/
/api/v1/jobs/
/api/v1/interviews/
/api/v1/flashcards/
/api/v1/chat/
/api/v1/documents/
/api/v1/progress/

24. Database Models

- User
- Profile
- Skill
- CareerGoal
- CareerAnalysis
- SkillGap
- LearningRoadmap
- RoadmapItem
- ProjectRecommendation
- UserProject
- Resume
- ResumeAnalysis
- JobDescription
- JobMatch
- InterviewSession

- InterviewQuestion
- InterviewAnswer
- InterviewFeedback
- FlashcardDeck
- Flashcard
- FlashcardProgress
- Document
- DocumentChunk
- ChatSession
- ChatMessage

25. Security

- Secure password storage
- JWT security
- CSRF protection for web
- Server-side authorization
- File type/size validation
- Private user documents
- AI endpoint rate limiting
- Secrets in environment variables
- Audit/error logging
- Never expose API secrets or internal prompts

26. Non-Functional Requirements

- Responsive web experience
- Mobile-ready APIs
- Modular agents
- Scalable long-running AI processing
- Database consistency
- Maintainable code
- Logging and monitoring
- Graceful AI failure handling

27. Development Roadmap

Phase 1 — Foundation

- Django project
- PostgreSQL
- DRF
- Authentication
- Profile
- React web shell

- API versioning
- Basic dashboard

Phase 2 — Core AI

- Career Agent
- Skill Gap Agent
- Learning Roadmap Agent
- Supervisor Agent

Phase 3 — Employability

- Project Agent
- Resume Analyzer
- Job Matching

Phase 4 — Learning & Practice

- Interview Agent
- AI Flashcards
- RAG
- AI Chat

Phase 5 — Quality

- Verification Agent
- Progress tracking
- API hardening
- Web/mobile synchronization

Phase 6 — Mobile

- React Native app
- Mobile authentication
- Dashboard
- Flashcard touch-flip
- Resume/document upload
- Interview UI
- Push notifications
- Android/iOS release preparation

28. Future Mobile Features

- Android and iOS
- Push notifications
- Offline flashcard caching
- Camera/document scanning
- Microphone interview practice
- Deep links
- App analytics

29. Complete Product Flow

Register → Login → Profile → Career Goal → Supervisor → Career Analysis + Skill Gap → Learning Roadmap → Projects → Resume → Job Matching → Interview → Flashcards → RAG/AI Chat → Verification → Personalized Career Plan → Progress Tracking → Web/Mobile Sync

30. MVP Success Criteria

- User can register and create a profile.
- Career analysis is generated.
- Skill gaps are identified and prioritized.
- Learning roadmap is generated and tracked.
- Relevant projects are recommended.
- Resume can be analyzed.
- Job description can be matched.
- AI interview can be completed.
- Flashcards can be generated and flipped.
- Uploaded documents can be queried through RAG.
- Progress persists in the backend.
- The same account and data can be used by web and future mobile clients.

31. Key Product Decision

API-first, backend-first architecture. Django + DRF owns authentication, business logic, AI orchestration, database access, RAG coordination and progress. React.js is the web client; React Native is the future mobile client. This avoids rebuilding the backend when the mobile app is introduced.

32. Final Product Definition

AI Multi-Agent Life & Career Copilot is a web-and-mobile-ready Agentic AI career platform where a Django backend coordinates specialized AI agents and RAG services, while React.js and React Native provide web and mobile experiences. The platform guides users from career selection and skill-gap discovery through learning, projects, resumes, job matching, interview preparation, AI flashcards and long-term progress tracking.